EGM305 Practical 1 – Classroom Carry Guide

Vertical Structure of the Atmosphere

Use this in the lab

This is the compact version: the calculation route, the physical reason behind each answer, the exact values you need, and a step-by-step plotting workflow. If a formula appears again, follow its live cross-reference back to where it is derived.

The whole practical as one chain

Q1 mass above halves with height → **Q2** pressure tracks overlying mass → **Q3** use the curve/formula at real heights → **Q4** oxygen partial pressure falls with total pressure → **Q5** temperature also changes with height.
Q6 plot real temperature profiles → **Q7** spot inversions → **Q8** locate tropopause → **Q9** calculate a lapse rate → **Q10** add tropical Florida → **Q11** infer changing tropospheric thickness.
Q12–14 explain pressure gradients, stability, and regional climate.

Three quantities that must not be mixed up

Density ρ: local mass in a local volume, $\rho = m/V$ (e.g. kg m^{-3}).
Percentage above: fraction of the *whole atmospheric mass* still above a chosen height.
Pressure p: force per unit area; in hydrostatic balance it reflects the weight per unit area of the air above.

Unit habits

1000 m = 1 km. Use km in the 5.6 km rule and in lapse rates expressed per km.
1 hPa = 100 Pa. Temperature differences have the same numerical size in K and °C.

Graph habits

For the sounding graph: **x-axis = temperature (°C)**, **y-axis = height (m or km)**. A table row (p, z, T) becomes the plotted point $(x, y) = (T, z)$. Pressure is *not* plotted on that graph.

Fast conceptual traps

- "Half every 5.6 km" means multiply the *remaining* amount by 1/2 each step; do not subtract 50 percentage points each time.
- Dry air at altitude is still about 21% oxygen by composition. What falls is oxygen *partial pressure*.
- An inversion is temperature *increasing* upward.
- "Cold air sinks" is only a local buoyancy shorthand; a descending parcel compresses and warms.
- The +3°C value derived from the 5.6-km rule is an *effective isothermal model temperature*, not the actual temperature at 5.6 km (about −21°C in the standard lapse-rate model).

Q1 Percentage of atmosphere above

What are you being asked?

Complete the halving table and plot percentage of atmospheric mass above against height.

At sea level take 100%. One 5.6-km rise leaves half:

$$100 \to 50 \to 25 \to 12.5 \to 6.25 \to \cdots$$

Why multiplication? Each new amount is the *old amount times one-half.*

After n halvings,

$$F = \left(\frac{1}{2}\right)^n. \tag{1.1}$$

Because each halving corresponds to 5.6 km,

$$n = \frac{z}{5.6}, \qquad z \text{ in km.} \tag{1.2}$$

Substitute Eq. (1.2) into Eq. (1.1) and multiply by 100:

$$\boxed{P(z) = 100\left(\frac{1}{2}\right)^{z/5.6}}. \tag{1.3}$$

The exponent $z/5.6$ literally means "how many 5.6-km halving intervals have I climbed?" It can be fractional: at 2.8 km you have climbed half of one halving interval.

Answer

0.0 km:	100.000%
5.6 km:	50.000%
11.2 km:	25.000%
16.8 km:	12.500%
22.4 km:	6.250%
28.0 km:	3.125%
33.6 km:	1.563%
39.2 km:	0.781%
44.8 km:	0.391%
50.4 km:	0.195%

If asked to solve backwards for height

Starting from Eq. (1.3),

$$\frac{P}{100} = 2^{-z/5.6}.$$

Take $\log_2$ of both sides:

$$\log_2(P/100) = -z/5.6.$$

Therefore

$$\boxed{z = 5.6\log_2\left(\frac{100}{P}\right) = 5.6\frac{\ln(100/P)}{\ln 2}}. \tag{1.4}$$

$\log_2(100/P)$ is simply the number of halvings required.

Q2 Pressure and why it decreases upward

Pressure is force per area:

$$p = \frac{F}{A}. \tag{2.1}$$

At a horizontal level, the atmosphere above has weight. "The layer supports the air above" does *not* mean a rigid shelf: pressure forces in the fluid balance gravity.

For a thin slab of area A and thickness Δz,

$$V = A\Delta z, \quad m = \rho A\Delta z,$$

and vertical force balance gives

$$p(z)A - p(z+\Delta z)A - \rho g A\Delta z = 0.$$

Divide by $A\Delta z$ and let $\Delta z \to 0$:

$$\boxed{\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g}. \tag{2.2}$$

The minus sign says p decreases when z increases. Integrating the same idea upward gives

$$p(z) = g\int_z^\infty \rho(z')\,\mathrm{d}z',$$

so pressure is the weight per unit area of the column above. For this practical, sea-level pressure is rounded to 1000 hPa, hence

$$\boxed{p = 1000\frac{P}{100} = 1000\left(\frac{1}{2}\right)^{z/5.6}\ \text{hPa}}. \tag{2.3}$$

So 50% above $\leftrightarrow$ 500 hPa, 25% $\leftrightarrow$ 250 hPa.

Where scale height comes from

Ideal gas law for dry air:

$$p = \rho R_d T \quad \Rightarrow \quad \rho = \frac{p}{R_d T}. \tag{2.4}$$

Put Eq. (2.4) into Eq. (2.2):

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{g}{R_d T}p.$$

If T is treated as constant, define

$$\boxed{H = \frac{R_d T}{g}}. \tag{2.5}$$

Taking reciprocals gives $1/H = g/(R_d T)$, hence

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\frac{p}{H}. \tag{2.6}$$

Separate and integrate:

$$\boxed{p(z) = p_0 e^{-z/H}}. \tag{2.7}$$

At $z = H$, $p/p_0 = e^{-1} = 0.368$: that is what *scale height* means. Half-pressure height follows from $1/2 = e^{-z/H}$:

$$z_{1/2} = H\ln 2. \tag{2.8}$$

If the lab fixes $z_{1/2} = 5.6$ km, then $H = 5.6/\ln 2 = 8.08$ km and

$$T = \frac{gH}{R_d} \approx 276\ \text{K} \approx +3°\text{C}.$$

+3°C versus −21°C

+3°C is the single constant temperature that makes an *imaginary isothermal atmosphere* halve exactly in 5.6 km. The standard troposphere instead cools from about 15°C at sea level to $15 - 6.5(5.6) \approx -21.4$°C at 5.6 km. They answer different questions.

Q3 Use the rule at real heights

Use Eqs. (1.3) and (2.3); convert metres to km first.

Place	z km	$z/5.6$	P %	p hPa
Portstewart	0	0	100.0	1000
Everest	8.848	1.580	33.4	334
Jet	12.000	2.143	22.6	226
Machu Picchu	2.430	.434	74.0	740
Balloon	21.027	3.755	7.4	74

Everest example:

$$P = 100(0.5)^{8.848/5.6} = 33.4\%,$$

$$p = 1000(0.334) = 334 \text{ hPa}.$$

Graph-reading version

Find the height on the vertical axis, move horizontally to the curve, then down/across to read P and pressure. The formula is the exact check when the graph reading is approximate.

Q4 Oxygen at altitude

Dry air is about 21% O_2. Dalton's law gives

$$\boxed{p_{O_2} = 0.21p}. \tag{4.1}$$

The worksheet's "effective oxygen percentage" is

$$\boxed{O_{2,\text{eff}}\% = \frac{p_{O_2}}{1000 \text{ hPa}} \times 100}. \tag{4.2}$$

Place	p hPa	p_{O_2} hPa	effective O_2 %
Portstewart	1000	210	21.0
Everest	334	70.1	7.0
Jet	226	47.5	4.7
Machu	740	155	15.5
Balloon	74	15.5	1.6

Watch this

Do not say Everest air is chemically "7% oxygen". Dry-air composition remains about 21%; its oxygen partial pressure is much lower.

Q5 Standard atmosphere and lapse rate

A lapse rate describes temperature decrease with height. The positive convention used here is

$$\boxed{\Gamma = -\frac{\Delta T}{\Delta z} = \frac{T_{\text{lower}} - T_{\text{upper}}}{z_{\text{upper}} - z_{\text{lower}}}}. \tag{5.1}$$

For the practical's standard troposphere,

$$\boxed{T(z) = 15 - 6.5z}, \quad z \text{ in km, to 11 km.} \tag{5.2}$$

Example, 6 km: $T = 15 - 6.5(6) = -24°\text{C}$.

z m	0	1k	2k	4k	6k	8k	10k	11k	14k	16k
T°C	15	8.5	2	-11	-24	-37	-50	-56.5	-56.5	-56.5

Q6 Plot the radiosonde profiles

Exactly what goes where

Each sounding row is (p, z, T). Plot T **horizontally** and z **vertically**. Example: Hillsborough (500 hPa, 5414 m, −22.45°C) becomes

$$\boxed{(x, y) = (-22.45, 5414)}.$$

The 500 hPa value is useful meteorological information, but it is *not an axis in Q6*.

Hand-plot recipe, in order

1. Label x-axis "Temperature (°C)", about −80 to +30.
2. Label y-axis "Height (m)", about 0 to 32,000 m (or 0–32 km if your sheet uses km).
3. Plot every Hillsborough (T, z) pair. Use one colour/symbol.
4. Connect *successive Hillsborough points in height order* with straight line segments.
5. Repeat separately for St John's with another colour/symbol.
6. Plot the standard-atmosphere points from Q5 as a third profile.
7. Add a legend. Do not connect one dataset to another.
8. Later, Q10 adds Florida to this same temperature-height framework.

Q6 data to plot – keep this page open beside your graph

Hillsborough, 21 March 1990

p hPa	z m	T °C	p hPa	z m	T °C
1001	38	11.05	853	1330	-1.25
850	1358	-1.25	835	1503	-1.05
735	2504	-7.45	573	4405	-15.25
549	4725	-18.05	500	5414	-22.45
460	6025	-25.45	400	7018	-32.85
282	9383	-51.05	250	10164	-49.05
216	11116	-44.65	150	13524	-49.65
120	14969	-53.85	100	16126	-56.45
70	18379	-58.65	50	20482	-58.45
30	23691	-60.05	20	26218	-56.25

St John's, 21 March 1990

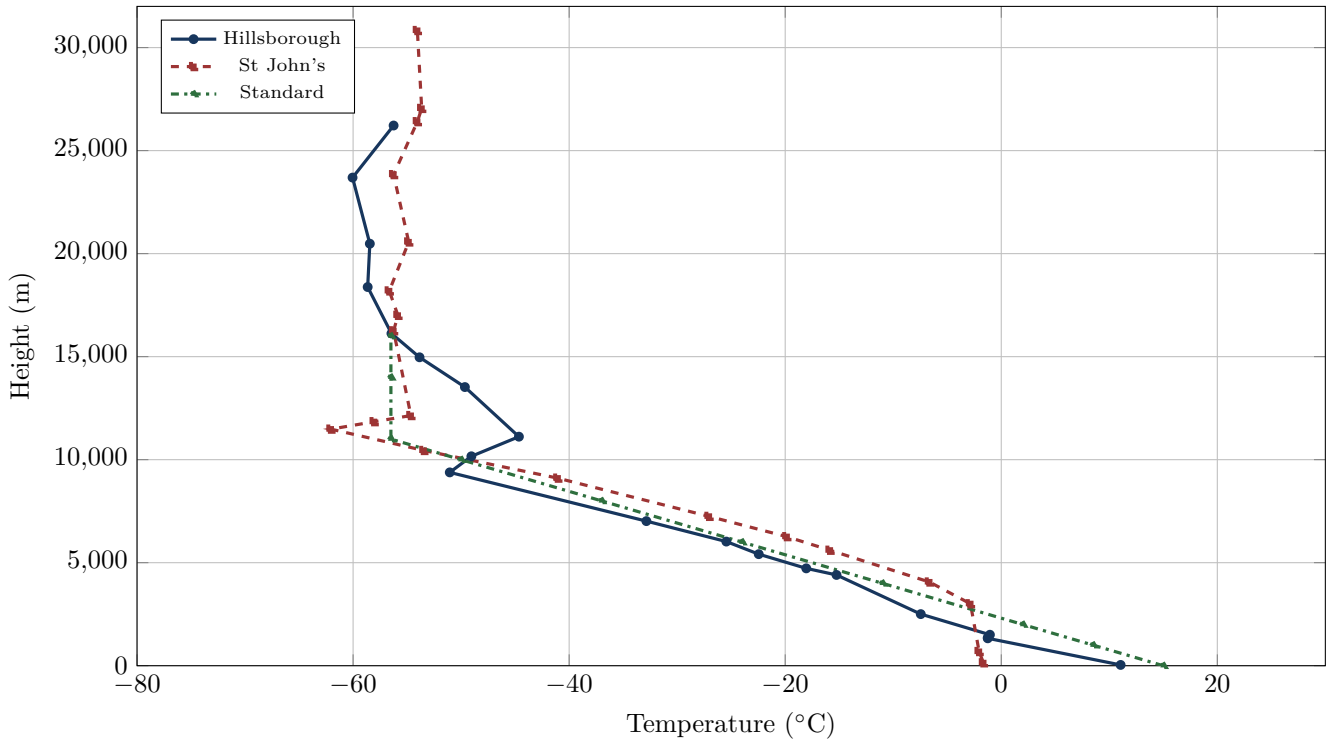
p hPa	z m	T °C	p hPa	z m	T °C
1005	140	-1.65	938	687	-2.05
700	3006	-2.85	612	4062	-6.65
500	5604	-15.85	458	6263	-19.85
400	7248	-27.05	307	9103	-41.05
250	10446	-53.45	212	11482	-62.05
200	11843	-58.05	191	12141	-54.65
100	16254	-56.25	88.9	17004	-55.85
73.6	18190	-56.65	50.5	20566	-54.85
30	23834	-56.25	20	26404	-54.05
18.1	27047	-53.65	10	30828	-54.05

How to read the shape once plotted

Going **up and left**: temperature is decreasing with height – normal tropospheric behaviour.
Going nearly **straight up**: temperature is nearly constant – an isothermal layer.
Going **up and right**: temperature increases with height – an inversion.
A marked change from strong cooling to nearly constant temperature or warming is evidence for the tropopause region.

Do not connect by pressure order accidentally

The table is already arranged from low to high altitude. Connect neighbouring rows of the *same station* in that height sequence. Pressure decreases upward automatically, but pressure is not the plotted coordinate.



Completed Q6 reference plot. Reproduce the profiles on the lab graph; the exact visual scale may differ from the handout.

Q7 Find inversion layers

An inversion satisfies

$$\Delta z > 0 \quad \text{and} \quad \Delta T > 0. \tag{7.1}$$

On a T-versus-height graph, the line therefore moves **right as it goes upward**.

Answer

Hillsborough: small inversion 1358–1503 m ($-1.25 \rightarrow -1.05$°C); strong upper inversion 9383–11116 m ($-51.05 \rightarrow -44.65$°C).
St John's: 11482–12141 m ($-62.05 \rightarrow -54.65$°C).
Florida later: 16000–18000 m ($-75.6 \rightarrow -73.3$°C).

Why stable? Lift a parcel into an inversion: surroundings become warmer above; the parcel tends to be colder/denser than its new environment and receives a restoring downward buoyancy force.

Q8 Locate the tropopause

The troposphere is the lower layer in which temperature generally decreases upward. The tropopause is the transition where that strong decrease stops, becomes weak, or reverses.

How to infer it from your graph

Start at the surface and trace the profile upward. Find the first high-altitude region where the persistent leftward cooling trend ends and the line becomes nearly vertical or bends right. Then use the original data table to read the corresponding height and temperature accurately.

Answer

Standard: about 11,000 m, −56.5°C.
Hillsborough: about 9383 m, −51.05°C.
St John's: about 11,482 m, −62.05°C.

Q9 Hillsborough mean tropospheric lapse rate

Use the surface and tropopause identified in Q8. From Hillsborough:

$$T_L = 11.05\text{°C},\ z_L = 38\ \text{m} = 0.038\ \text{km},$$

$$T_U = -51.05\text{°C},\ z_U = 9383\ \text{m} = 9.383\ \text{km}.$$

Apply Eq. (5.1):

$$\Gamma = \frac{11.05 - (-51.05)}{9.383 - 0.038} = \frac{62.10}{9.345} = \boxed{6.65\text{°C/km}} \approx 6.6.$$

Use positive 6.6°C/km when reporting the conventional lapse-rate magnitude. If using $\Delta T/\Delta z$ directly, the gradient would be negative because temperature falls upward.

Q10 Add Florida

Plot these as (T, z) and connect in order:

z m	T°C	z m	T°C
0	27.2	10000	-32.6
2000	15.2	12000	-50.5
4000	6.1	14000	-64.5
6000	-6.7	16000	-75.6
8000	-17.9	18000	-73.3

Cooling continues to 16 km and then reverses, so

Answer

Florida tropopause $\approx 16,000$ m at ≈ -75.6°C. (If an answer sheet says -75.4°C, the table itself supports -75.6°C.)

Q11 Tropospheric thickness

Your graph provides the evidence: Hillsborough ~ 9.4 km, St John's/standard ~ 11 km, Florida ~ 16 km. Therefore it is **not the same thickness everywhere**.

For two pressure surfaces, the hypsometric relation is

$$\Delta z = \frac{R_d \bar{T}}{g} \ln\left(\frac{p_1}{p_2}\right). \tag{11.1}$$

For fixed p_1, p_2, a warmer mean column $\bar{T}$ gives a larger Δz: warm columns are geometrically deeper. Tropical convection and latent heating also extend vigorous tropospheric mixing higher.

Answer

Troposphere is generally deepest in the tropics (roughly 16–18 km) and shallowest toward the poles (roughly 8–9 km).

Q12 Why the pressure drop differs with height

St John's lower interval:

$$1005 - 700 = \boxed{305 \text{ hPa}}, \quad 3006 - 140 = 2866 \text{ m}.$$

Upper interval:

$$30 - 20 = \boxed{10 \text{ hPa}}, \quad 26404 - 23834 = 2570 \text{ m}.$$

The height spans are similar but Eq. (2.2) says

$$\left|\frac{\mathrm{d}p}{\mathrm{d}z}\right| = \rho g.$$

Near the surface, ρ is high: each vertical metre contains much more mass, so pressure loses many hPa per km. High up, ρ is very small, so the absolute hPa change is small.

Yet the fractional drops are similar:

$$305/1005 \approx 30\%, \qquad 10/30 \approx 33\%.$$

That is consistent with exponential-like pressure decay: similar vertical intervals remove similar *fractions*, not similar absolute hPa.

Q13 Why cold upper air does not simply sink

"Cold air sinks" only works when comparing parcels at approximately the *same pressure*. Upper-tropospheric air is cold partly because it is at low pressure and has expanded.

If a parcel descends, surrounding pressure increases, so it is compressed and warms adiabatically. For dry unsaturated air the dry adiabatic lapse rate is about

$$\Gamma_d = \frac{g}{c_p} \approx 9.8^\circ\text{C/km}. \tag{13.1}$$

Buoyancy asks whether the displaced parcel is denser or lighter than the *environment at its new level*. In a stable atmosphere, displacement produces a restoring force.

A useful pressure-corrected temperature is potential temperature:

$$\theta = T\left(\frac{p_0}{p}\right)^{R_d/c_p}. \tag{13.2}$$

For dry static stability, $\mathrm{d}\theta/\mathrm{d}z > 0$.

Answer

Upper air cannot fall to the surface while staying at its original cold temperature: descent compresses and warms it. Stability is determined by parcel-vs-environment density at the same level, not by comparing surface temperature directly with upper-air temperature.

Q14 Hillsborough versus St John's

Surface sounding values:

$$11.05^\circ\text{C} \quad \text{versus} \quad -1.65^\circ\text{C},$$

so Hillsborough was

$$11.05 - (-1.65) = \boxed{12.70^\circ\text{C}}$$

warmer that day, despite being farther north.

The ~ 100 m station-height difference is too small to explain 12.7°C. The relevant controls are regional air masses and ocean currents: Northern Ireland has strong maritime influence from the North Atlantic and prevailing westerlies; St John's is affected by the cold Labrador Current and can receive very cold continental air from North America.

Answer

The contrast is not fixed all year. It tends to be especially strong in the cool season; seasonal land heating means St John's can become comparably warm or warmer in summer.

In-class "why?" answers

Why does pressure correspond to mass above?

For a hydrostatic column, the upward pressure force pA balances the column weight mg, so $p = mg/A$. More mass above per unit area means more pressure.

Why does surrounding pressure decrease as a parcel rises?

At a higher level there is less atmospheric mass remaining above that level. Hydrostatic balance therefore requires a smaller pressure there; locally, Eq. (2.2) gives the decrease per metre.

Why do fractional powers make sense?

$z/5.6$ need not be an integer. A half-step is represented continuously: $(1/2)^{1/2} = 1/\sqrt{2} \approx 0.707$. Exponential decay smoothly interpolates between the full halving levels.

Why do logs appear when solving for height?

Height is in the exponent in Eq. (1.3); logarithms are the inverse operation of exponentiation, so they "bring the exponent down" and let you solve for z.

Why does cooling upward matter for clouds?

As rising air expands and cools, saturation vapour pressure falls. If water-vapour amount does not fall as quickly, relative humidity rises; at the dew point saturation is reached and condensation can begin.

One-page classroom formula + answer sheet

Core equations

$$P(z) = 100(1/2)^{z/5.6} \quad \text{Eq. (1.3)}$$

$$p(z) = 1000(1/2)^{z/5.6}\ \text{hPa} \quad \text{Eq. (2.3)}$$

$$\frac{\mathrm{d}p}{\mathrm{d}z} = -\rho g \quad \text{Eq. (2.2)}$$

$$p = \rho R_d T \quad \text{Eq. (2.4)}$$

$$H = R_d T/g \quad \text{Eq. (2.5)}$$

$$p = p_0 e^{-z/H} \quad \text{Eq. (2.7)}$$

$$z_{1/2} = H \ln 2 \quad \text{Eq. (2.8)}$$

$$p_{O_2} = 0.21p \quad \text{Eq. (4.1)}$$

$$\Gamma = -\Delta T/\Delta z \quad \text{Eq. (5.1)}$$

$$T_{std} = 15 - 6.5z \quad \text{Eq. (5.2)}$$

$$\Delta z = \frac{R_d \bar{T}}{g} \ln(p_1/p_2) \quad \text{Eq. (11.1)}$$

Constants: $g = 9.81$ m s^{-2}, $R_d \approx 287$ J kg^{-1} K^{-1}, $c_p \approx 1005$ J kg^{-1} K^{-1}.

Q1–Q5 answers

Q1: 100, 50, 25, 12.5, 6.25, 3.125, 1.56, 0.781, 0.391, 0.195%.
Q2: 1000, 500, 250, 125, 62.5, 31.25, 15.6, 7.8, 3.9, 2 hPa.
Q3: Portstewart 100%,1000; Everest 33.4%,334; jet 22.6%,226; Machu 74.0%,740; balloon 7.4%,74.
Q4: $p_{O_2} = 210$, 70.1, 47.5, 155, 15.5 hPa; worksheet-effective $O_2 = 21.0$, 7.0, 4.7, 15.5, 1.6%.
Q5: at 0,1,2,4,6,8,10,11 km: 15, 8.5, 2, -11, -24, -37, -50, -56.5°C.

Q6–Q10 graph workflow

Q6: x = T, y = height. Plot each station separately; connect neighbouring points in height order. Pressure is not an axis.
Q7: rightward-with-height segment = inversion.
Q8: tropopause where persistent tropospheric cooling stops/weakens/reverses. Standard 11 km; Hillsborough 9.383 km; St John's 11.482 km.
Q9: Hillsborough mean lapse rate = 6.65 ≈ 6.6°C/km.
Q10: Florida tropopause ≈ 16 km, −75.6°C.

Q11–Q14 answers

Q11: Troposphere not equally thick; deepest tropics, shallowest poles. Warm columns expand and tropical convection reaches higher.
Q12: St John's drops 305 hPa low down versus 10 hPa high up over similar distances because ρ is much larger below.
Q13: cold upper air does not simply fall; descending parcels compress/warm, and buoyancy compares parcel with environment at the same pressure level.
Q14: Hillsborough 11.05°C vs St John's -1.65°C: North Atlantic maritime influence/westerlies versus Labrador Current/continental cold; seasonal contrast varies.

Last check before answering

- Did I convert m → km where needed?
- Am I using percentage *above*, not local density?
- Did I put temperature on x and height on y for Q6–10?
- Did I keep each profile separate when connecting dots?
- Am I reading an inversion as warming upward?
- Am I reporting lapse-rate magnitude as positive when temperature decreases upward?